

Lecture 13

Implementing Programming Languages

What is a functional programming language?

- Pure
- Immutable
- Expression oriented
- First Class functions
- Variants / Pattern matching

What is a functional programming language?

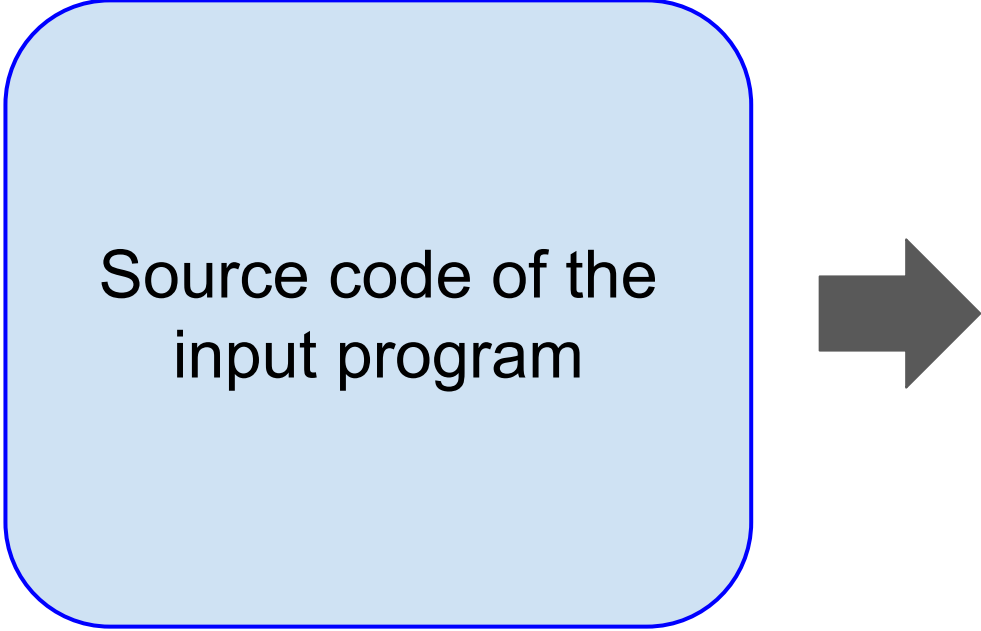
- Pure
- Immutable
- Expression oriented
- First Class functions
- Variants / Pattern matching

Some of these features exist in imperative languages as well!

e.g.

- lambdas in JS, Python, Java, etc
- immutability by default in Rust
- pattern matching in Scala, Rust, etc
- first class functions in JS, Ruby, Lua

What is a programming language?

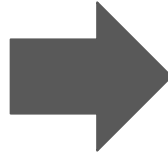


Source code of the
input program

A light blue rounded rectangle with a blue border contains the text 'Source code of the input program'. To its right is a large, solid dark gray arrow pointing to the right.

What is a programming language?

Source code of the
input program

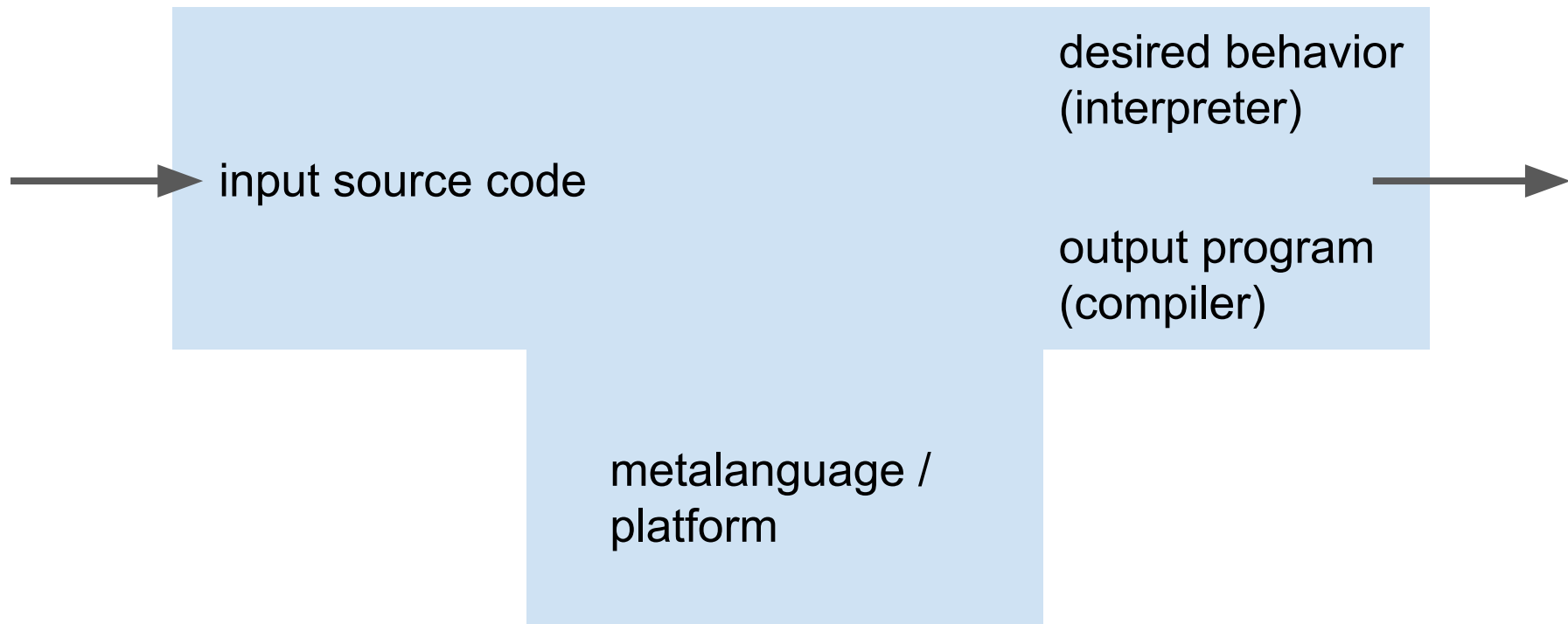


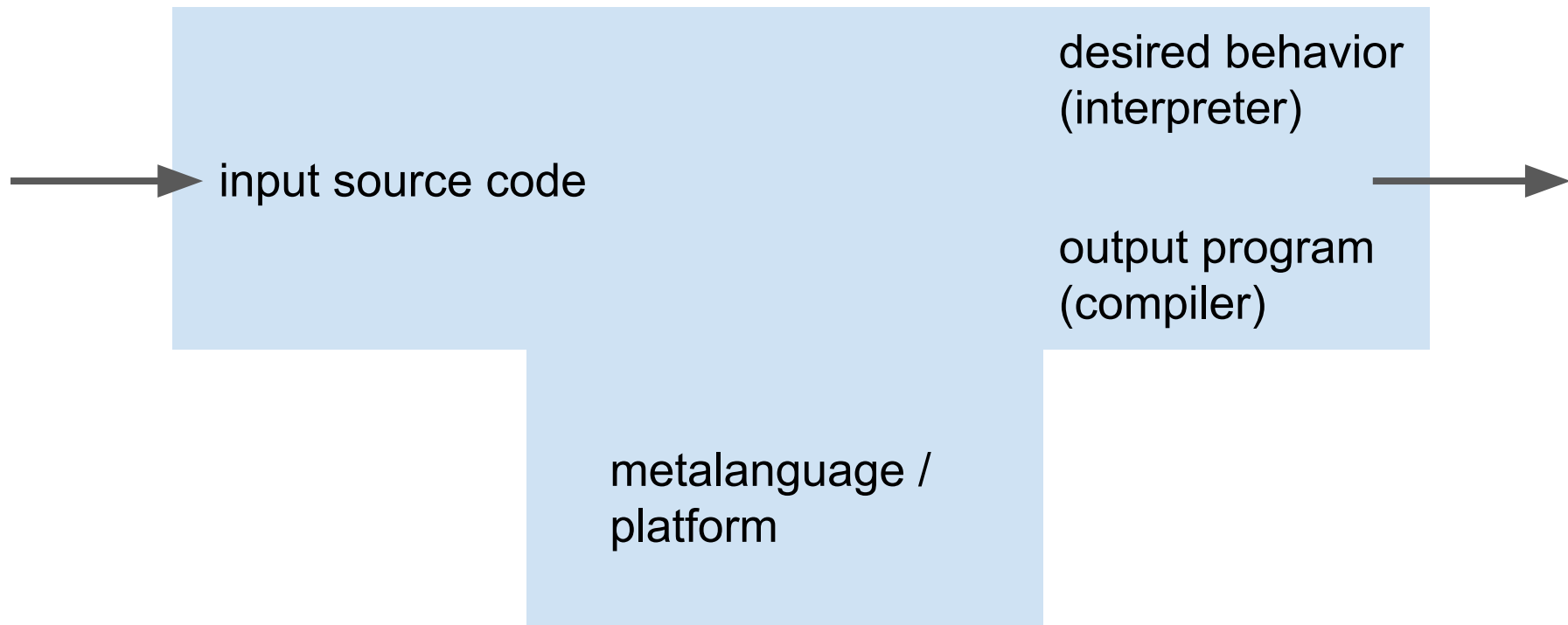
Desired Behavior:

- computed result
- stuff on the screen
- side effects
- etc

Two Approaches

- Compile: Translate to an existing language
 - CSE 401
- Interpret: Run program directly
 - CSE 341

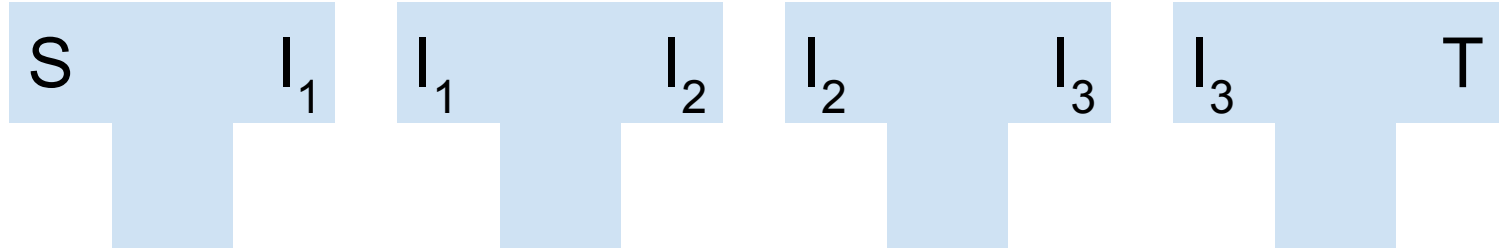




gcc : C (.c) $\rightarrow$ x86 (.s) $\rightarrow$ machine code (.o) $\rightarrow$ desired behavior

java : .java $\rightarrow$.class $\rightarrow$ desired behavior

Chaining Compilers Together



There's no such thing as a “compiled language”

- Interpretation or compilation is an **implementation detail**
- For any language, could write an interpreter or a compiler
- Many languages have one “default implementation”
 - Some languages have several implementations!

Phase 2: Trefoil

ThReE

Forty

One

Instructional

Language

Trefoil V0

A very simple programming language with four kinds of expressions:

1. `hw` prints `"hello world"`
2. `pi` prints the first 5 digits of pi
3. `echo e` prints the string `e`
4. `square e` prints the square of `e` (assuming `e` is a number)

Trefoil V0

What's missing?

- numbers
- lists + data structures
- more expressions
 - numeric operations
- variables
- functions
- recursion

Trefoil V0

A very simple programming language with four kinds of expressions:

- | | |
|--------------------------|---|
| 1. <code>hw</code> | prints <code>"hello world"</code> |
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| 3. <code>echo e</code> | prints the string <code>e</code> |
| 4. <code>square e</code> | prints the square of <code>e</code> (assuming <code>e</code> is a number) |

What's missing?

- | | | |
|---------------------|-------------------|--------------------|
| • multiple commands | • input/output | • standard library |
| • variables | • data structures | • loops |
| • functions | • recursion | • memory |
| • comments | • types | • and more! |

Group Work: Trefoil v0.5

Syntax:

x y where x and y are integers

And has the following semantics:

x y should evaluate to $x + y$

Examples:

`interpret "3 5"` evaluates to 8

`interpret "-1 -1"` evaluates to -2

Hints

- Your interpreter should be a function `interpret` that takes as input a `string` (the Trefoil v0.5 program text) and produces an `integer`
- `String.split_on_char ' ' s` will split `s` on spaces
- `int_of_string x` will return an integer corresponding to the integer value of `x` (and will throw an error if `x` is not parseable as an integer)
- Use `failwith "msg"` to raise an error if anything goes wrong